UNIVERSAL PLASTIC SHREDDER V2.0

Electrical Control & Power System

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The Problem

Ichor Systems generates large volumes of thick rigid plastic (PVC, PP, LDPE) during semiconductor equipment manufacturing. Off-the-shelf shredders are either too small for the material (0.25–0.5 in. sheets, 4–12 in. wide) or priced far above the sponsor's per-line budget. The EE team designed and built the electrical control and power system for an industrial-grade plastic shredder; the mechanical structure is delivered by a separate ME team.

Requirements

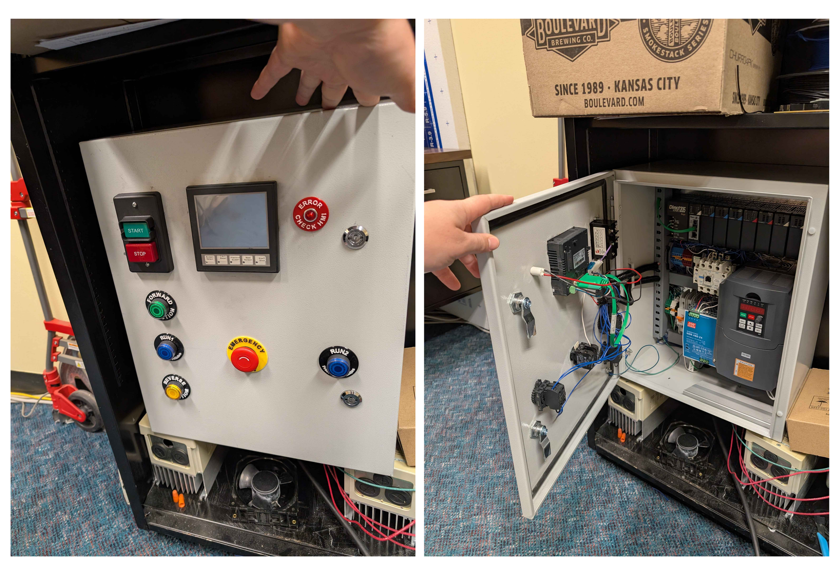
Must:

- Shred plastic to ∼1 in. pieces.
- Drive a 3–5 HP motor at 70–90 RPM through a controllable VFD.
- Follow NFPA 70 (NEC); include E-stop, overload, and one safety interlock.
- Provide an HMI for operator control and status.
- Stay within the $1,000 EE-team budget.

Should: status / fault lights, NEMA enclosure, labeled controls.
May: read-only wireless monitoring, data logging, capacitive-touch safety.

System Architecture

The control system is built end-to-end on AutomationDirect industrial hardware – no hobby-grade microcontrollers are in the safety or control path. A PowerTec 71008 disconnect feeds a Mean Well NDR-480-24 24 VDC supply, a Mitsubishi SD-N35 contactor (E-stop hardwired in series with its coil), and a Huanyang HY02D211B-T VFD. The VFD drives a 3 HP bench motor (GE 5KE182BC205B) over U/V/W via blue, white, and red 4 mm banana sockets.



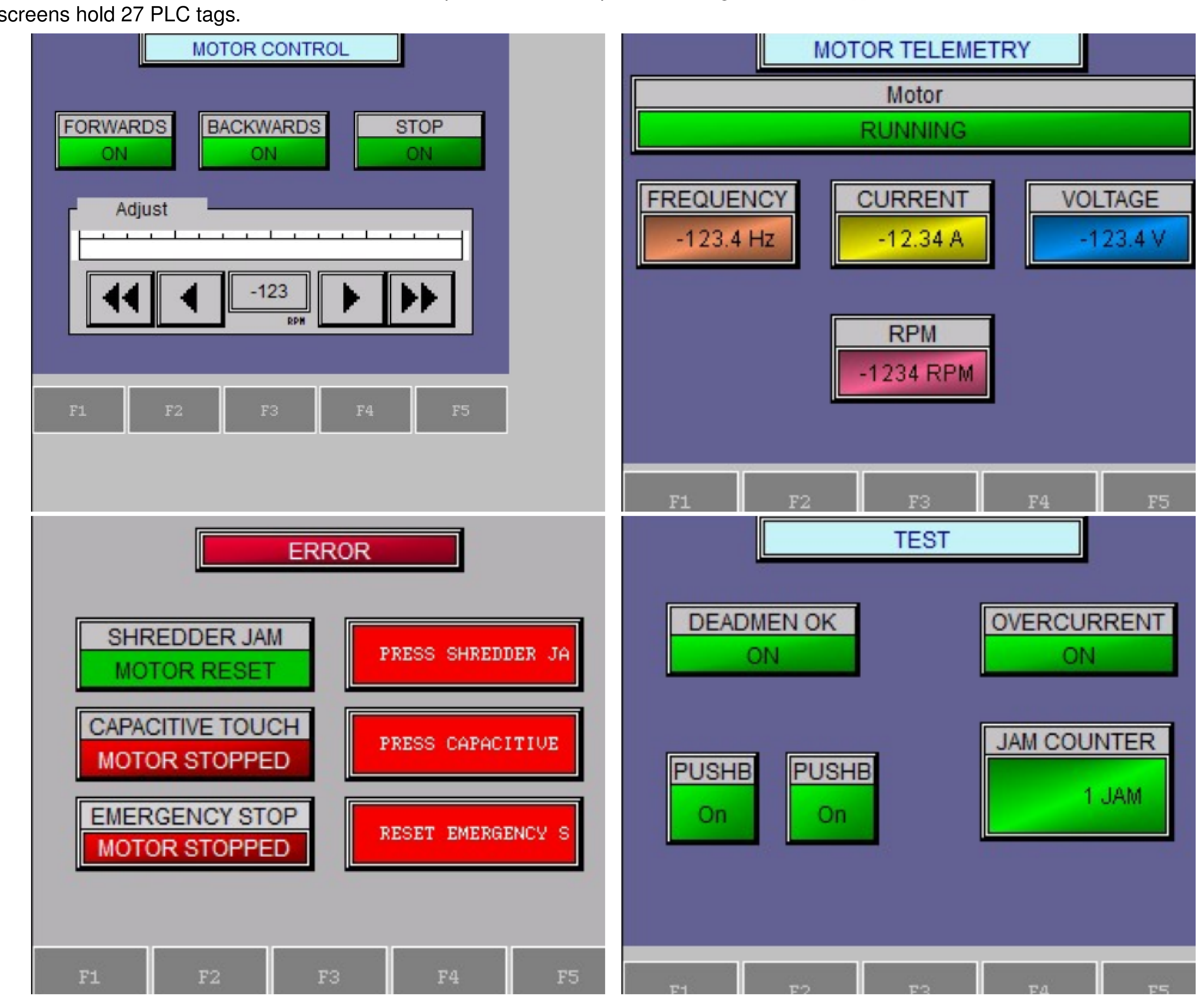
Operator face (left), panel interior (right): HMI, deadman / fwd / rev / E-stop, DirectLOGIC 205, Huanyang VFD, 24 V supply, SD-N35 contactor, terminal blocks.

PLC Control

The PLC is the system controller. A DirectLOGIC 205 D2-09B-1 rack holds the H2-DM1E CPU and five I/O modules: `D2-08ND3` (8 DC inputs), `F2-08AD-1` (analog current input), `F2-04RTD` (RTD), `D2-08TR` (8 relay outputs), and `F2-08DA-2` (0–10 V analog output to the VFD). The 31-rung Do-more program implements a deterministic state machine plus a 3-strike jam recovery sequence and a hard-lockout path.

HMI

AutomationDirect EA1-T4CL C-more Micro-Graphic 4 in. touch panel running `HMI.413.mgp`, serial to the H2-DM1E CPU. Four screens hold 27 PLC tags.



Motor Control, Motor Telemetry, Error / Alarm, Test.

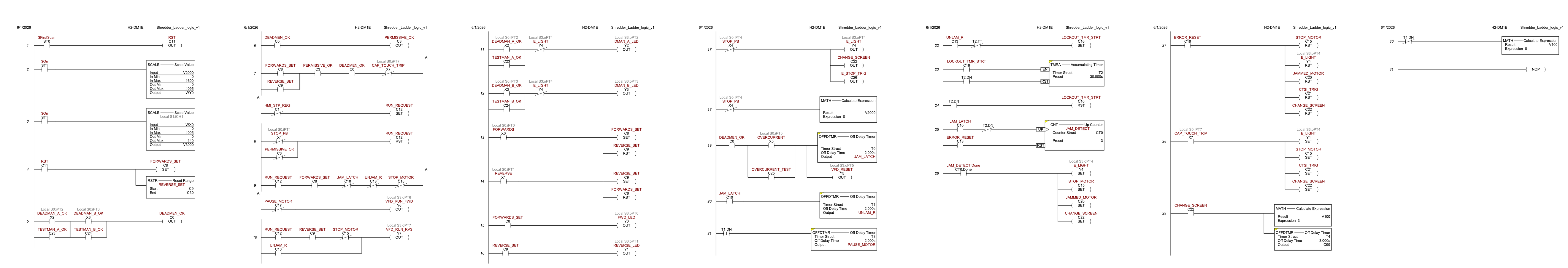
Safety & Compliance

- **E-stop:** Category 0 stop. Eaton E22B1 NC mushroom in series with the SD-N35 contactor coil. Drops the contactor independently of the PLC.
- **Two-hand / deadman:** both pushbuttons must be held to enable the motor (PLC X2 AND X3).
- **Overcurrent / jam:** VFD over-torque trip (`PD124 = 150%`, `PD125 = 3 s`) on PLC X5; ladder runs the 3-strike unjam routine.
- **Enclosure:** Vorksmen SEB001 steel, NEMA Type 12.
- **Compliance:** NFPA 70/79, NEMA 250 / ICS / MG 1, ISO 12100, 13849-1 (Cat B–1 / PL b–c), 13850, 13851, OSHA 1910.147.

Results

- Panel built, wired, bench-validated. All 31 rungs program and scan correctly.
- All four HMI screens operate; tag database matches the PLC.
- Speed reference verified: HMI `V2000` → `SCALE` → `F2-08DA-2 CH1` → VFD `VI`.
- Jam-recovery verified: overcurrent on `X5` → 2 s reverse (`T1`) → coast-to-stop buffer (`T3`).
- Three-revision point list signed off by Fearghus Tyler.

Complete Ladder Logic — 31 rungs across 7 pages



Pages 1–7: power-up init / deadman • RUN_REQUEST gating / Y6 Y7 outputs • indicator LEDs / direction latches • E-stop / jam detection / unjam stroke • lockout timer / jam counter • fault reset / cap-touch / HMI screen change • screen reset / end of program.

Project By the Numbers

31 ladder rungs in production.
27 HMI tags across 4 screens.
14 component sheets in the point list.
3 strikes before the motor locks out.
2 s unjam reverse stroke.
150% over-torque threshold on the VFD.

Hardware Stack

PLC: DirectLOGIC 205 + H2-DM1E.
HMI: C-more EA1-T4CL.
VFD: Huanyang HY02D211B-T.
Motor: GE 5KE182BC205B (3 HP bench).
PSU: Mean Well NDR-480-24.
Contactor: Mitsubishi SD-N35.

Future Work

Production motor selection (3–5 HP) supersedes `PD141--PD144`.
Mechanical integration and shred-load run pending ME team.
Capacitive-touch (CTSI) module on `X7`: machine-level verification pending.
Optional Modbus link to the VFD for richer telemetry.

Acknowledgements

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